

GradeAI — Official Answer Key

Subject: Science | Class: 10 | Exam: Practice Test | Total Marks: 50

Instructions for AI Grading:

Grade each question strictly based on the answer key. Award partial marks for partially correct answers.

Q1 (5 marks): What is photosynthesis?

Model Answer:

Photosynthesis is the process by which green plants use sunlight, water, and carbon dioxide to produce oxygen and energy in the form of glucose. The equation is: $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. It occurs in the chloroplasts of plant cells.

Q2 (5 marks): Define Newton's Second Law of Motion.

Model Answer:

Newton's Second Law states that the acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass. Formula: $F = ma$, where F is force in Newtons, m is mass in kg, and a is acceleration in m/s^2 .

Q3 (10 marks): Explain the structure of DNA.

Model Answer:

DNA (Deoxyribonucleic Acid) is a double helix structure consisting of two strands wound around each other. Each strand is made of nucleotides containing a sugar (deoxyribose), a phosphate group, and a nitrogenous base (Adenine, Thymine, Guanine, Cytosine). Base pairing rules: A pairs with T, G pairs with C. DNA carries genetic information and is found in the nucleus.

Q4 (10 marks): What are the differences between mitosis and meiosis?

Model Answer:

Mitosis: produces 2 identical daughter cells, maintains chromosome number (diploid), occurs in somatic cells, used for growth and repair. Meiosis: produces 4 genetically different cells, reduces chromosome number by half (haploid), occurs in reproductive cells, used for sexual reproduction. Key difference: meiosis involves two divisions (Meiosis I and II) while mitosis involves one.

Q5 (10 marks): Explain the water cycle.

Model Answer:

The water cycle involves: 1) Evaporation - water from oceans/lakes evaporates due to solar energy. 2) Transpiration - plants release water vapor. 3) Condensation - water vapor cools and forms clouds. 4) Precipitation - water falls as rain/snow. 5) Collection - water collects in oceans, rivers, lakes, and groundwater. The cycle continuously recycles water through the environment.

Q6 (10 marks): Describe the structure and function of the human heart.

Model Answer:

The human heart has 4 chambers: right atrium, right ventricle, left atrium, left ventricle. Right side pumps deoxygenated blood to lungs (pulmonary circulation). Left side pumps oxygenated blood to body (systemic circulation). Valves (tricuspid, bicuspid, pulmonary, aortic) prevent backflow. Heart rate averages 72 beats/minute. Controlled by sinoatrial node (pacemaker).